R6.7: the ζ-fold of the Gaudry base over $\mathbb{F}_{p^2}$, phase two

Round 6, 2026-10-18 (run start 20261018T2056Z UTC), base commit abee07ee73. Example `examples/r6_7_zeta_fold_phase2.rs`; commands in `run.sh`; raw records `*.json` in this directory; full tables in `RESULTS.md`. Shared cloud container, seven runs concurrent, no isolation; wall times contended and unused. Toy synthetic curves only; no claim about rho or deployed curves.

Arms

At $m = 3$ on the torsion family: **A** plain $F = \{x \in \mathbb{F}_p^*\}$ folded by $\langle -1, \zeta \rangle$; **B** T-saturated $F_c = \varphi^{-1}(F'')$ on E (18 points a column), run through a wrapper around the repository `LineOracle`; **C** the quotient F'' on E'. Cost unit: $\mathbb{F}_p$ multiplications, inversions counted and priced at the Fermat cost I_p (12 to 25). The counting copy of the solver matched the repository solver on every call; B lift failures: 0.

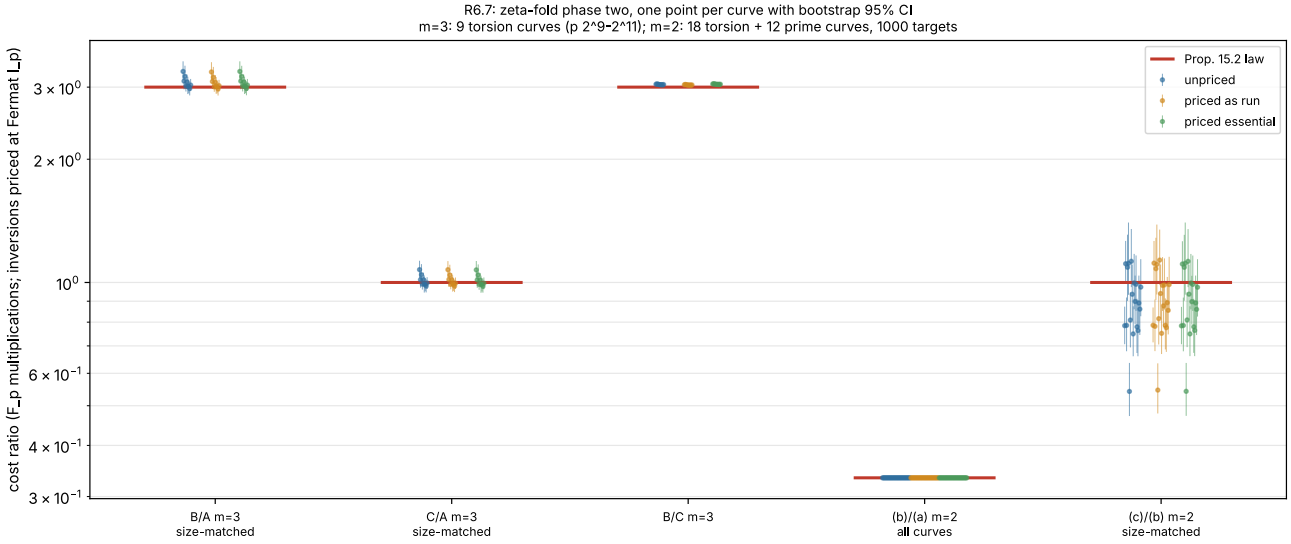


Figure 1: Measured cost ratios, one point per curve with bootstrap 95% CI over targets (1000 reps), in three pricings; red = Prop. 15.2 law. $m = 3$: 9 torsion curves, $p \approx 2^9$ to 2^{11}, 100 to 200 targets. $m = 2$: 18 torsion and 12 prime curves, 1000 targets. Source: `m3_*.json`, `m2_*.json`.

ratio	law	measured (range over curves)	CIs containing law
B/A size-matched, $m = 3$	3	2.976 to 3.277 (priced as run 2.969 to 3.267)	8 of 9
C/A size-matched, $m = 3$	1	0.980 to 1.075	8 of 9
B/C, $m = 3$	3	3.037 to 3.055 (CI width < 0.001)	0 of 9
(b)/(a), $m = 2$	1/3	0.3333 on 30 of 30	all, every pricing
(c)/(b) size-matched, $m = 2$	1	0.542 to 1.127	12 of 18

Inversions are 38 to 44 % of the priced cost in every arm, so pricing moves no ratio by more than 0.011 at $m = 3$ or 0.024 at $m = 2$. B/C exceeds 3 by about 1.4 % on every curve: this is the wrapper's own charged work (φ and the lift), which the law does not count.

Relation streams

Prime family (9 curves): full rank and log recovered on all 9, (a)/(b) relations to full rank 2.79 to 4.97. Torsion family (9 curves): base exhausted before (a) reaches full rank on all 9, (b) reaches full rank on 3 of 9. The round-4 stall is reproduced with the repository oracle.

Limits

$p \leq 2^{11}$ at $m = 3$, $\leq 2^{16}$ at $m = 2$; 3 seeds per size; Fermat inversion price, not a hardware I/M.